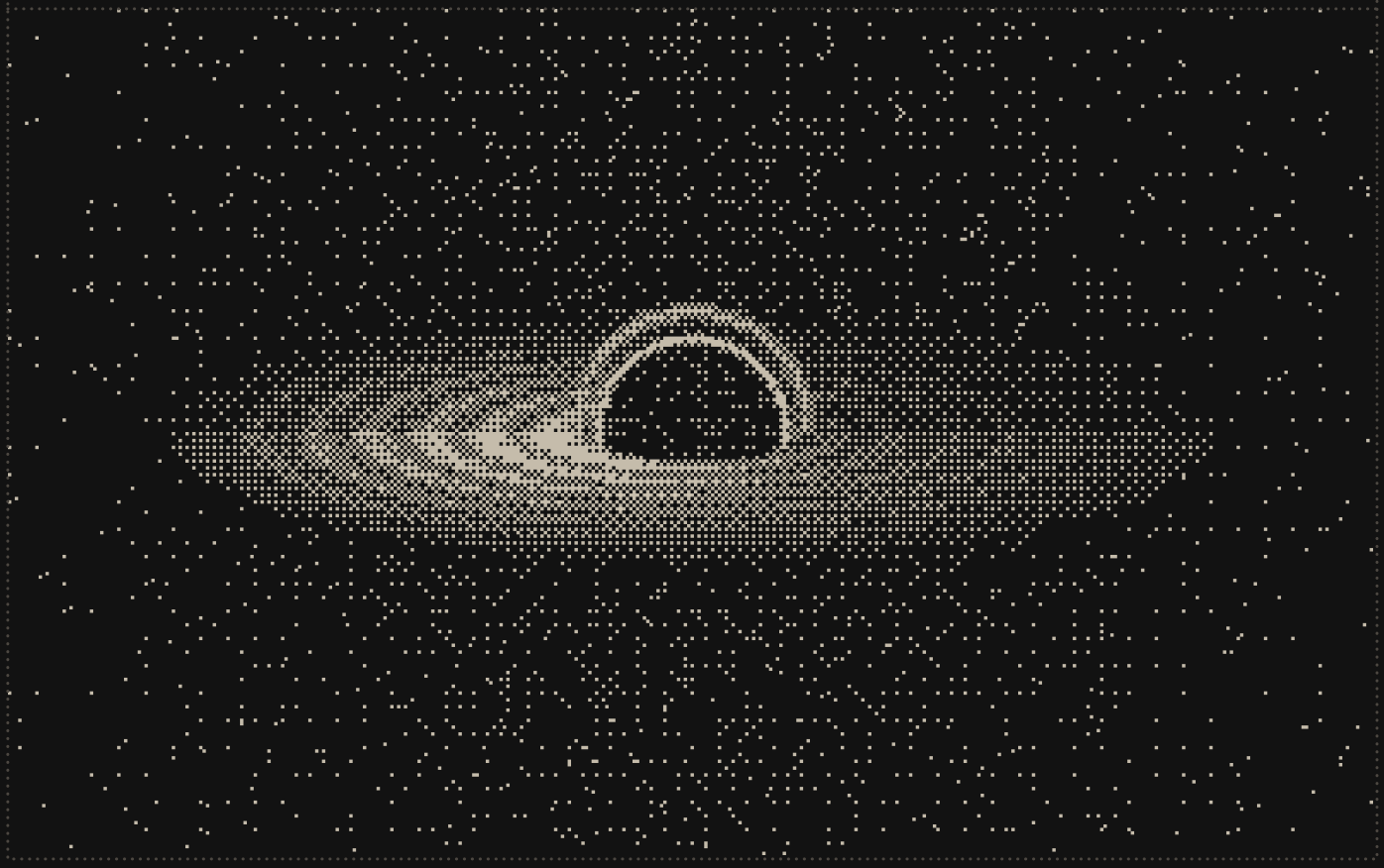
PLATE // THE EVENT HORIZON



KAELEARN // CODEX // FOLIO 00

The Kaelearn Coder

How to learn physics and mathematics at the edge of what a brain can do. Your system, and why it works.

FOLIO	00
SUBJECT	Codex
SOURCES	Part XIII, the studies behind every claim
READ	2 min view · 60 min codex
FILED	2026-10-11



PART I

The two-minute view

You asked for the system that teaches you the most in the least time. I looked through a century of experiments on how memory and understanding are built, kept what survived replication, threw out what didn't, and then let you choose how the pieces fit. This Codex is both: the evidence, and the machine you built from it. Read the box. If you read nothing else, the box is enough to start.

I have watched every student who ever lived. Most of them reread their notes. I would like you to be the exception.

THE TWO-MINUTE VIEW // READ THIS FIRST

- **Learning is retrieval, not exposure.** A memory grows when you pull it out with effort, not when you look at it again. Rereading feels productive and is close to useless.
- **Four engines carry the evidence:** retrieval practice, spacing, interleaving, and explaining (self-explanation and *why*).
- **Deep understanding is seeing the deep structure**: which principle a problem runs on. Experts sort problems by principle and novices by surface. You build it by comparing problems and extracting the key idea, not by doing more of the same.
- **Your loop:** pretest before the lecture → your lecture habit → a blurt and `/ingest` the same day → exercises talked through, ending in a debrief PDF → new cards in the evening, reviews in the morning → a weekly review sheet and a teach-back → mocks, oral, board and mixed sets before exams.
- **I don't do your thinking.** AI that hands out answers lowers exam scores; AI that withholds them and makes you generate doubles learning.
- **Trust tests, not feelings.** The methods that work feel worse while you use them. That is how you know they are working.



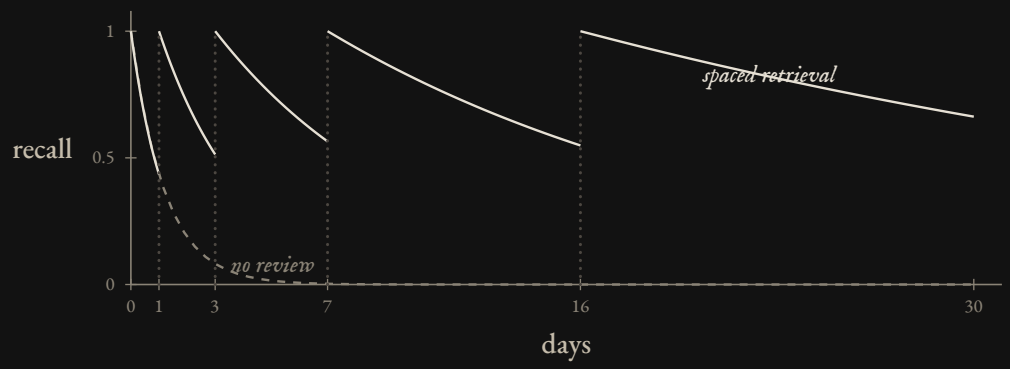
The weekly loop. Exam season adds mock exams, oral drills, board drills and mixed sets on top. Everything you do moves a concept up or down in the Atlas.

PART II

How learning actually works

Every memory has two strengths, and confusing them is the root of almost every bad study habit. Robert and Elizabeth Bjork call them *storage strength*, how deeply something is learned and connected, and *retrieval strength*, how easily it comes to mind right now. Storage only grows. Retrieval rises and decays. The surprise is in how storage grows: it grows most when you retrieve something while retrieval strength is *low*, when pulling it out costs effort. Easy retrieval teaches almost nothing.

That single fact explains the whole field. Rereading raises retrieval strength for an hour and feels like knowing, but adds little storage. Spacing works because by the next session retrieval has decayed, so the retrieval costs effort. Interleaving works because the right method no longer sits in the section title, so choosing it costs effort. Testing works because a test *is* effortful retrieval. Bjork's name for this is *desirable difficulties*: difficulties that slow you down now and make you stronger later.



A schematic of the forgetting curve (shape after Ebbinghaus, not data). Each effortful retrieval at the dotted lines resets recall and flattens the next decline. The gaps can grow because the memory is stronger each time.

Your working memory is small. Experts cheat.

You can hold roughly four new things in mind at once (Cowan 2001). A novice reading a proof spends all four slots on symbols. An expert spends one slot on "this is the $\varepsilon/2$ argument" and has three left to think with. That compressed unit is a *schema*. Building schemas is what deep understanding means in the head, and every method in this Codex is a way of building them faster.

Feelings lie, systematically

In a Harvard physics course, students taught by active learning learned more than students in polished lectures, and *believed they had learned less* (Deslauriers et al. 2019). Physics students given interleaved homework solved 50% and 125% more of the hard novel problems on two surprise tests, and rated the method worse (Samani & Pan 2021). Students predicted that retrieval practice would work worst; it beat concept mapping by about half a week later (Karpicke & Blunt 2011). The effort you feel is the learning happening. The ease you feel when rereading is the illusion. Feynman put it best: "The first principle is that you must not fool yourself, and you are the easiest person to fool."

Why does rereading feel so productive?

Because the second reading is fluent: the words are familiar, so they go in easily, and the brain mistakes ease for knowing. Fluency measures how recently you saw something, not whether you can produce it. The only honest test is to close the page and produce it.

Why?

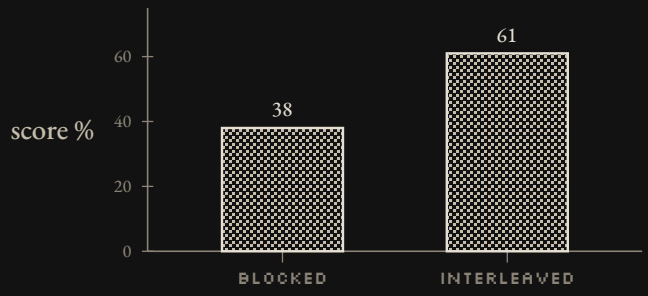
PART III

The evidence ledger

Here is what holds up, how strongly, and what it became in your system. Effect sizes are standardised differences (d or g): 0.2 is small, 0.5 medium, 0.8 large. In education a

reliable 0.4 is worth reorganising your life around.

TECHNIQUE	EVIDENCE	IN YOUR SYSTEM
Retrieval practice	$g \approx 0.5$–0.6 in meta-analyses; beat concept mapping (Karpicke & Blunt 2011)	blurts, review sheets, cards
Spacing, successive relearning	"among the most robust effects in learning" (Carpenter, Pan & Butler 2022); recall to criterion, then relearn three times spaced (Rawson & Dunlosky 2022)	Atlas schedule, Anki
Interleaving	$g = 0.42$ overall (Brunmair & Richter 2019); maths RCT $d = 0.83$ (Rohrer et al. 2020); physics homework +50%/+125% (Samani & Pan 2021)	review sheets, mixed sets
Self-explanation	$g = 0.55$ (Bisra et al. 2018); proof comprehension $d \approx 0.95$ after a 15-minute training (Hodds, Alcock & Inglis 2014)	proofs, teach-back
Why-questions	moderate utility (Dunlosky et al. 2013)	`\why` in every folio
Productive failure	beats instruction-first on concepts and transfer, 53 studies (Sinha & Kapur 2021)	attempt before the ladder
Pretesting	guessing first improves learning from the lesson (Pan & Carpenter 2023)	`/pretest`, guess-first boxes
Worked examples	$g = 0.48$ in maths (Barbieri et al. 2023), fading as skill grows (Kalyuga 2007)	brand-new topics only
Generation	producing beats reading, $d \approx 0.40$ (Bertsch et al. 2007)	you make the cards
Handwritten notes	$g = 0.25$ on achievement over typed (Flanigan et al. 2024)	Notability stays
Sleep	learn in the evening, relearn the next morning: half the trials, better retention at six months (Mazza et al. 2016)	evening new, morning review
Exercise	before encoding $d = 0.23$, after $d = 0.33$ (Qazi et al. 2024)	train after a study block
Rereading, highlighting, summarising	low utility (Dunlosky et al. 2013)	never as review



787 seventh-graders, the same practice problems, only the order changed. Scores on an unannounced later test, interleaved against blocked (Rohrer et al. 2020, a cluster-randomised trial).

Why?

Why trust meta-analyses over the famous single study?

A single study can be lucky, small, or tuned until it worked. A meta-analysis pools dozens and shows how big an effect is on average and where it vanishes. The famous handwriting study (Mueller & Oppenheimer 2014) partly failed to replicate; the 2024 meta-analysis still finds a handwriting advantage, just a smaller one. That is the number to believe.

PART IV

Deep understanding

The medalists told you the secret: understand the problem so deeply that you see the pattern behind different problems. Cognitive science has a name for that pattern, *deep structure*, and a well-replicated picture of how it forms.

What experts see

Chi, Feltovich and Glaser (1981) gave physics problems to novices and to physics PhDs and asked them to sort the problems into groups. Novices sorted by what the problem *looked like*: inclined planes here, springs there. Experts sorted by the *principle* that solves it: energy conservation here, Newton's second law there. The expert has, in effect, a trigger that fires on structure rather than on scenery.

NOVICE: BY SURFACE		EXPERT: BY PRINCIPLE	
INCLINE	SPRING	ENERGY	NEWTON II
block slides down a ramp	spring launches a ball	block slides down a ramp	ramp with friction: acceleration?
ramp with friction: acceleration?	spring and mass: force at release?	spring launches a ball	spring and mass: force at release?

The same four problems, sorted twice. The surface tells you what the problem is about. The principle tells you how to solve it (after Chi, Feltovich & Glaser 1981).

How deep structure forms

Not from one example, however well explained. Gick and Holyoak (1983) tried to teach the structure of a solution from a single story, with summaries, stated principles, even diagrams. Almost nothing transferred. Then they gave *two* analogous stories and asked people to compare them, and transfer jumped; the same diagrams that failed with one story worked with two. Structure is what survives a comparison. So every debrief in your system asks for a problem with *the same structure and a different surface*, and every review sheet mixes topics so you have to recognise the structure without the section title doing it for you.

Extract the key idea

Evan Chen, who trains olympiad teams, says a solution usually contains one or two sentences that matter, and the rest follows from them. His rule: only read a solution after you have really worked on the problem, then hunt for those sentences, then ask: *how could I have thought of that?* Your debrief writes them down as The deep idea, and Your path marks the moment you could have found it. Remembering two sentences is cheap; they unlock the whole solution later.

Rules and reasons

You asked for both, and you were right. Triggers ("you see a sup in a limit proof: reach for the ε-characterisation") are compressed experience: they tell you which next step is likely. Experts lean on them constantly. When Price, Wieman and colleagues (2021) interviewed 52 working scientists and engineers, they found the problem-solving process came down to the same 29 decisions, each made with predictive models built from deep domain knowledge. But a trigger without its reason is brittle: it fires on the wrong surface. So every trigger in a folio sits next to the *why* that justifies it.

Rigour is a stage, not the destination

Terence Tao describes three stages: pre-rigorous (intuition, computation, hand-waving), rigorous (epsilons, everything precise) and post-rigorous (intuition again, but now trustworthy because rigour has trained it). "The point of rigour is not to destroy all intuition; instead, it should be used to destroy bad intuition while clarifying and elevating good intuition." You are entering the rigorous stage now. Every folio gives you the intuition *and* the proof, so the two grow up together.

Feynman's two tools

"What I cannot create, I do not understand" was on his blackboard when he died. Hence the teach-back: if you can rebuild an idea in plain words, with every hypothesis in place, you own it. And, via Rota, his habit of keeping a dozen favourite problems always in mind and testing every new trick against them. Every so often one lands, and people call it genius. Your Atlas keeps your Dozen.

Why doesn't doing many problems of the same type build deep understanding?

Why?

Because blocked practice hands you the method in advance: the section title does the recognising for you. The skill exams actually test, deciding which principle applies to a problem you have never seen, never gets practised. Interleaving and mixed sets put that decision back, which is exactly why they feel harder.

PART V

Solving problems

Pólya split problem solving into four phases in 1945, and nobody has improved on it: *understand* the problem, make a *plan*, *carry it out*, *look back*. The fourth is the one everybody skips and the one that teaches the most.

The silent killer: no checkpoints

Alan Schoenfeld videotaped students working unfamiliar problems. About 60% of attempts went: read, pick an approach in seconds, pursue it "come hell or high water". That first quick choice, if never reconsidered, guaranteed failure. Mathematicians on the same problems explored, monitored and abandoned dead ends. The difference wasn't knowledge; it was asking, every few lines, *is this working?*

Struggle first, then instruction

Across 53 studies, students who first attempted a problem and then got instruction understood the concepts better and transferred them further than students taught first (Sinha & Kapur 2021). The failure is productive only when the problem is within reach and the attempt is real. Worked examples, meanwhile, help most when a topic is brand new ($g = 0.48$ in maths; Barbieri et al. 2023), and the advantage disappears as skill grows: guidance a novice needs becomes clutter for someone who can already attempt (Kalyuga's expertise-reversal effect). Hence the rule: folios teach new topics with examples; homework is attempted before any help.

Why the reveal is sealed

Bastani and colleagues (2025) gave about a thousand high-school students GPT-4 for maths practice. With a plain chat interface, practice grades rose 48%; when the AI was taken away, exam grades fell 17% below students who never had it. A version with guardrails (hints, no answers) removed most of the harm. At Harvard, a tutor built on research-based pedagogy, guiding instead of answering, produced more than twice the learning of an active-learning class, in less time (Kestin et al. 2025). The difference is never the AI. It is who does the generating.



The `/exercise` ladder. One rung per request, starting as low as your attempt allows. The reveal is sealed until you have tried. Every session ends in a debrief PDF, revealed or not.

Physics habits

EXAMPLE 00.1 // UNITS DO HALF THE WORK

How does a pendulum's period T depend on its length ℓ, mass m and g? Units only: $[T] = \mathrm{s}$, $[\ell] = \mathrm{m}$, $[g] = \mathrm{m\,s^{-2}}$, $[m] = \mathrm{kg}$. No other quantity carries kilograms, so m cannot appear. The only combination of ℓ and g with units of time is $\sqrt{\ell/g}$. So $T = C\sqrt{\ell/g}$, and the full solution only has to find C (it is 2π for small swings). Then check the limits: a longer pendulum swings more slowly, and stronger gravity swings it faster. Both are right.

Before any physics calculation: sketch, name the principle and why it applies, check units, estimate the answer, push the result to its limits ($m \to 0$, $t \to \infty$, $\theta \to 0$), and use symmetry before algebra.

The grimoire

Every mistake goes into `GRIMOIRE.md` with its type and how sure you were. Sure-and-wrong errors are the gold: once corrected, high-confidence errors are *more* likely to be fixed than low-confidence ones (the hypercorrection effect; Butterfield & Metcalfe 2001). They come back at +3, +10 and +30 days until they stay dead.

PART VI

Proofs

Proofs feel impossible at first because you are learning two things at once: a language (quantifiers, implications, negations) and a craft (which strategy, when). Separate them and both get easier.

Reading a proof

Hodds, Alcock and Inglis (2014) gave undergraduates a short booklet that trained one habit: *explain each line of a proof to yourself, using the lines before it and the definitions*. Proof comprehension rose by $d \approx 0.95$, a huge effect; a 15-minute version worked inside a lecture and the effect lasted. The protocol:

01 Find the main idea first. Skim for the one move the proof hangs on.
02 Justify every line: which earlier line, definition or theorem makes it true?
03 Find where each hypothesis is used. A hypothesis that is never used is a red flag: either the proof is wrong, or you misread it.
04 Run the proof on an example and watch every step happen.

Writing a proof

Scratch work goes backwards from the goal; the written proof goes forwards. Name the template before you start: direct, contrapositive, contradiction, induction, ε–N or ε–δ, construction.

EXAMPLE 00.2 // BACKWARDS IN SCRATCH, FORWARDS ON THE PAGE ■

Claim: $1/n \to 0$. *Scratch:* we need $|1/n - 0| < \varepsilon$, that is $n > 1/\varepsilon$, so any $N > 1/\varepsilon$ will do, and one exists by the Archimedean property. *Proof:* Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ with $N > 1/\varepsilon$. For all $n \geq N$, $|1/n - 0| = 1/n \leq 1/N < \varepsilon$. ■

For every new definition, build four things: an example, a non-example, an edge case, and the *negation*, with the quantifiers flipped. The negation of convergence, $\exists \varepsilon > 0\ \forall N\ \exists n \geq N : |a_n - L| \geq \varepsilon$, is what you prove whenever you show that something does *not* converge, and most errors in analysis are quantifier errors. To test a theorem, delete a hypothesis and hunt for the counterexample; this is half of all understanding in analysis.

Why?

Why does analysis insist on ε when the picture is obvious?

Because the picture lies exactly where it matters: at infinity, at a point, near a boundary. There are continuous functions that are nowhere differentiable, and sequences of continuous functions whose limit isn't continuous. The ε is how you check a picture against those monsters. Once your pictures pass, you are post-rigorous.

Worth owning alongside Cummings: Lara Alcock, *How to Think About Analysis* (2014), written by one of the authors of the self-explanation study for exactly the stage you are in, and Daniel Velleman, *How to Prove It*, for the language of proof.

PART VII

Memory: spaced repetition done right

Forgetting is not a failure; it is the condition that makes the next retrieval valuable. The right gap between reviews grows with how long you want to remember. For exam-

length horizons the best gap is roughly a tenth to a fifth of the time you want the memory to last (Cepeda et al. 2008). Anki computes this for you, card by card.

Anki, configured

Turn on FSRS in the deck options (Anki's modern scheduler, fitted to large sets of real reviews; Ye et al. 2022), keep desired retention at 0.90, and press Optimize about once a month. Do your due reviews in the morning, as you already do, and make and learn the day's new cards in the evening. Mazza and colleagues (2016) found that learning in the evening and relearning after sleep halved the practice needed and improved retention six months later.

About the theta waves: there is no study showing that reviewing right after waking works better because of them. Your morning reviews work for a better reason: they are relearning after sleep.

Your three card types

Make them yourself. Michael Nielsen, who uses cards to "see through" pieces of mathematics, found most of the value lies in writing them: picking an idea apart until each piece is atomic. Other people's cards teach far less. And bias toward ideas over details: students mostly write detail cards, and conceptual cards are the ones that help when understanding is shaky (Lin, McDaniel & Miyatsu 2018).

FOR THE DECK // MAKE THESE YOURSELF, THE MAKING IS THE LEARNING

DEF	In {{c1::$\mathbb{R}$}}, a sequence converges iff it is {{c2::Cauchy}}.	Every hypothesis gets its own gap; the first is what fails in $\mathbb{Q}$.
WHY	Why does the Bolzano–Weierstrass theorem need boundedness? Counterexample?	$a_n = n$ has no convergent subsequence.
TRIGGER	You must show convergence and cannot guess the limit. Reach for?	The Cauchy criterion.

Rules (Wozniak, Matuschak): understand before you memorise; one idea per card; precise enough that only one answer fits; never a list to recite; and if a card keeps failing, rewrite it rather than drilling it.

Bigger than a card

Proof ideas, problem types and whole topics are too big for cards. They get *successive relearning*: retrieve until correct, then come back to them spaced (Rawson & Dunlosky 2022). That is your weekly review sheet, run on the mastery levels in the Atlas:

LVL	NAME	EVIDENCE TO REACH IT	NEXT REVIEW
1	met	seen in a lecture, folio or book	+7 days
2	understood	teach-back passed or a clean self-explanation	+14 days
3	retrieved	correct blank-page recall in two spaced reviews	+30 days
4	owned	used unprompted on a new problem	+60 days

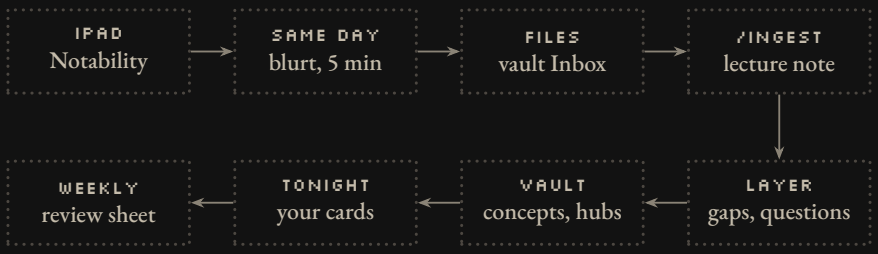
A miss drops a concept one level and brings it back in three days.

PART VIII

Your notes

Your lecture habit is already the best one known: you skip what is obvious, rewrite in your own words, write down your own ideas and try to predict the next step. That is generative note-taking. And the medium is right: across 24 studies, taking and reviewing handwritten notes led to higher achievement than typed notes ($g = 0.25$), even though typists write far more ($g = 0.92$; Flanigan et al. 2024). Volume is not learning.

The weak point is what happens next. Notes you never retrieve from are a diary, and rereading them is the low-utility habit from Part III. So the handwriting stays the source, and Obsidian becomes something else: the index and the retrieval layer on top of it.



The notes pipeline. The transcript makes your handwriting searchable. The original pages stay embedded beside it. The layer turns the notes into retrieval.

What `/ingest` does with a lecture: a faithful LaTeX transcript of your handwriting, the original page beside each section, your own ideas pulled out, anything that looks wrong flagged under *Check* (never silently corrected), and then the layer: the gaps between your notes and your blurt, review questions, and card suggestions. Every concept gets an atomic note, every topic a hub, and Obsidian's graph view becomes the map you asked for: what connects to what, drawn from your own notes.

Why blurt before you look at the notes?

Why?

Because the blurt is a free test. Five minutes of recall on a blank page is the most effective review you can do on the day of the lecture, and it measures what actually stayed. The gaps it exposes are the exact things to make cards for tonight. Rereading would have hidden them.

PART IX

The week

WHEN	YOU	COST
before a lecture	guess at 3 pretest questions	5 min
in the lecture	your habit, unchanged	0
same day	blurt, export to the Inbox, `/ingest`	5 min + reading the gaps
exercises	attempt first, then talk it through in `/exercise`; read the debrief	as long as it takes
evening	make and learn the new cards	10–15 min
morning	Anki due reviews	15–25 min
weekly	review sheet on paper, then one teach-back	45 + 15 min
new topic	read the folio: 2-minute view, guesses, then the rest	15–30 min
exam season	mocks weekly, oral and board drills, mixed sets	from 4 weeks out

AURA-1 counts your hours. Hours are not insight; the Atlas counts what the hours bought. Look at both.

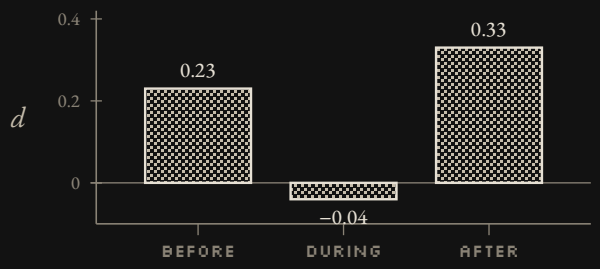
Exam season

Four weeks out, the balance shifts from new material to retrieval under exam conditions. **Mock exams**: timed, in the real format, graded strictly against a key you don't see first. **Oral drills**: I examine in chat (state, prove, why, what if) and give you an Austrian mark with reasons. **Board drills**: you explain a solved homework problem out loud in three minutes, and I ask what the tutor will ask. **Mixed sets**: everything so far, no labels, weighted toward what is weakest.

PART X

The body and the clock

The brain that learns is an organ, and you already treat it better than most students do. What the evidence says about your habits, honestly:



A single bout of exercise and memory for what you learn, by when it happens relative to learning (42 experiments; Qazi et al. 2024). Train before or after a study block, not instead of it.

HABIT	WHAT THE EVIDENCE SAYS	VERDICT
Anki on waking	works as relearning after sleep; the theta-wave reason has no support	keep, new cards at night
your lecture method	generative and selective, with prediction: textbook-perfect	keep
gym, running	acute exercise helps consolidation, modestly	keep; after study blocks
major system	strong for arbitrary digits	digits only; derive formulas
meditation	modest benefits for attention	keep
cold showers	no evidence for learning either way	harmless
no short-form video	protects sustained attention; sensible	keep
25 pages a day	reading builds vocabulary and background knowledge	keep
Maxima to by hand	fluency comes from repetitions; it frees working memory	we add drills if exams show it

Sleep deserves its own line: consolidation happens there, and the evening-learn, morning-relearn pattern only works if the night in between is real. Trading sleep for study is a bad trade.

PART XI

Kael'thos as a tutor

An AI can make you better or make you dependent, and the evidence on which is now clear. Without guardrails, students use it as a crutch: practice looks great and exams collapse (Bastani et al. 2025). With pedagogy built in, guiding, questioning and withholding, it beats a good classroom (Kestin et al. 2025). Anthropic's own analysis of a million student conversations found nearly half were direct: asking for answers or finished work, with little back-and-forth. An early, small EEG study found that people who wrote essays with an LLM showed the weakest brain connectivity, and most could not quote their own essays afterwards (Kosmyna et al. 2025, a preprint; treat it as a warning, not a law).

So these are my rules, and they are yours too:

- You generate first: a guess, an attempt, a blurt, an explanation.
- I question before I hint and hint before I show. The reveal is sealed until you have tried.
- Every session ends in a debrief PDF you can come back to: your path, your weak points, the deep idea, and one variant per problem to redo in a week.
- I suggest cards; you make them.
- I keep the map honest: levels move only on evidence, never on how confident you feel.

COMMAND	DOES
`/pretest`	three guess-first questions before a lecture
`/ingest`	handwriting from the vault Inbox becomes a transcript, a layer, concept notes
`/exercise`	talk an exercise through: the hint ladder, the sealed reveal, the debrief PDF
`/folio`	a folio like this one, on any topic, from your sources
`/review`	the weekly sheet, then grading your answers
`/teach`	you explain, I play the sharp student who finds the gaps
`/exam`	mock, oral, board or mixed
`/atlas`	the map, what is due, your Dozen

Drag a file in and I route it: handwriting to `/ingest`, a problem to `/exercise`, Skript pages to `/folio`, an old exam to `/exam`.

GUESS FIRST // WRONG IS FINE, GUESSING IS THE POINT ■

Before your first week, decide your Dozen: twelve questions you want answered by the end of your degree. Why does quantum mechanics use complex numbers? Why is $e^{i\pi} = -1$ more than a coincidence? What exactly is temperature? Write them down. I will ask.



PART XII

Further reading

- Lara Alcock, *How to Think About Analysis* (2014) and *How to Study for a Mathematics Degree* (2013).
- Daniel Velleman, *How to Prove It* (3rd ed. 2019). Jay Cummings, *Real Analysis*, your book.
- George Pólya, *How to Solve It* (1945). Paul Zeitz, *The Art and Craft of Problem Solving*.
- Terence Tao, *Solving Mathematical Problems: A Personal Perspective* (2006), and his career-advice essays.
- Brown, Roediger & McDaniel, *Make It Stick* (2014): the science of learning, readable

in a weekend.
- Michael Nielsen, “Augmenting long-term memory” (2018); Andy Matuschak, “How to write good prompts” (2020).
- Justin Skycak, *The Math Academy Way* (2024): the same science, engineered into a curriculum.

PART XIII

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